

The input-referred noise power spectral density is then

$$V_{in\text{noise}}^2(f) = \frac{V_{onoise}^2(f)}{A_v^2} = \left(\frac{g_{m2}}{g_{m1}}\right)^2 \cdot \left(\frac{1}{g_{m2}}\right)^2 \cdot (I_{M1}^2 + I_{M2}^2) \quad (21.31)$$

Minimizing the input-referred noise is accomplished by making g_{m1} large. Using the expressions in Eq. (9.66), we also see that minimizing g_{m2} reduces the input-referred noise. This is equivalent to saying large A_v results in minimum input-referred noise.

21.1.2 The Source Follower (Common-Drain Amplifier)

Source-follower configurations using a gate-drain-connected load are shown in Fig. 21.15. The drain of the amplifying device, that is, the MOSFET *not* gate-drain-connected (the load), is common to both the input and the output, so this topology is often called a common-drain amplifier. Setting the bias current in this topology can be challenging. In the following we don't concern ourselves with this issue.

A source follower implemented in CMOS has an asymmetric drive capability; that is, the ability of the follower to source current is not equal to the ability to sink current for a given bias condition and AC input signal. The small-signal gain of the NMOS source follower shown in Fig. 21.15 is simply determined by a voltage divider between the resistance looking into the source of M2 ($1/g_{m2}$) with the resistance of the gate-drain-connected load, M1 ($1/g_{m1}$). The gain of the **source follower amplifier** is

$$A_v = \frac{v_{out}}{v_{in}} = \frac{\text{resistance connected to the source}}{\text{resistance connected to the source} + \text{resistance looking into the source}} \quad (21.32)$$

or for the NMOS source follower in Fig. 21.15

$$v_{in} = \overbrace{v_{gs2}}^{i_d/g_{m2}} + \overbrace{v_{out}}^{i_d/g_{m1}} \text{ where } (v_{gs1} = v_{out}) \text{ and so } v_{out} = v_{in} \cdot \frac{\frac{1}{g_{m1}}}{\frac{1}{g_{m1}} + \frac{1}{g_{m2}}} \quad (21.33)$$

The output resistance of the source follower shown in this figure is

$$R_{out} = \frac{1}{g_{m1}} \parallel \frac{1}{g_{m2}} \quad (21.34)$$

Note that M2 can only source current, while M1 can only sink current, and the gain of the source follower is always less than one.

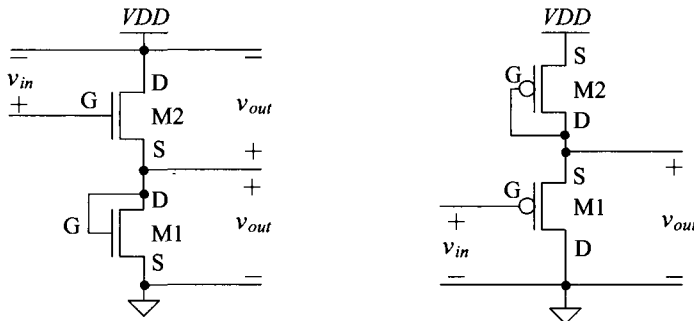


Figure 21.15 Source followers (common drain) using gate-drain loads.

21.1.3 Common Gate Amplifier

The common gate amplifier with gate-drain load is shown in Fig. 21.16. The input resistance of this amplifier is simply the resistance looking into the source of M1, or

$$R_{in} = \frac{1}{g_{m1}} \quad (21.35)$$

The gain of this **common-gate amplifier** is

$$A_v = \frac{v_{out}}{v_{in}} = \frac{-i_d \cdot \frac{1}{g_{m2}}}{-v_{gs1}} = \frac{-i_d \cdot \frac{1}{g_{m2}}}{-i_d \cdot \frac{1}{g_{m1}}} = \frac{\text{resistance in the drain}}{\text{resistance in the source}} = \frac{\frac{1}{g_{m2}}}{\frac{1}{g_{m1}}} \quad (21.36)$$

or noting the positive gain

$$A_v = \frac{g_{m1}}{g_{m2}} = \sqrt{\frac{KP_n}{KP_p} \cdot \frac{W_1 L_2}{W_2 L_1}} \quad (21.37)$$

which is the same form as the gain for the CS amplifier with gate-drain load.

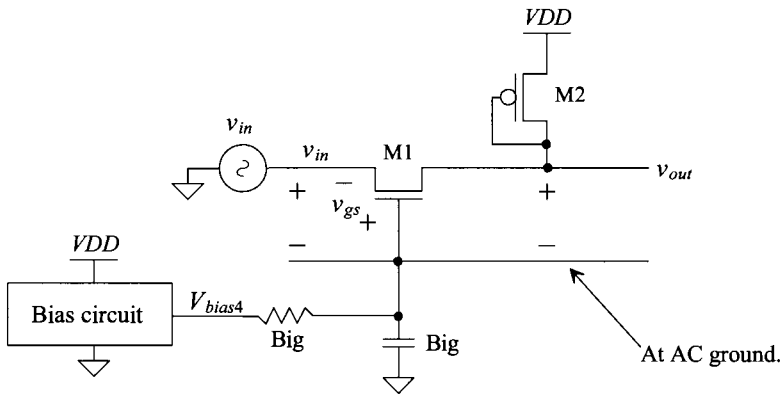


Figure 21.16 The common gate amplifier.

21.2 Current Source Loads

In the last section we talked about the gate-drain MOSFET part of a current mirror. In this section we talk about the current-source side of the mirror. The current source load provides an amplifier with the largest possible load resistance and thus the largest gain.

21.2.1 Common-Source Amplifier

Consider the CS amplifier with current source load shown in Fig. 21.17. The MOSFET M1 is the common-source component of the amplifier, while the MOSFET M2 is the current source load. The DC transfer characteristics are also seen in this figure. The amplifying portion of the curve occurs when the output is not close to V_{DD} or ground (both M1 and M2 are saturated). The slope of the line when both transistors are saturated corresponds to the gain of the amplifier.

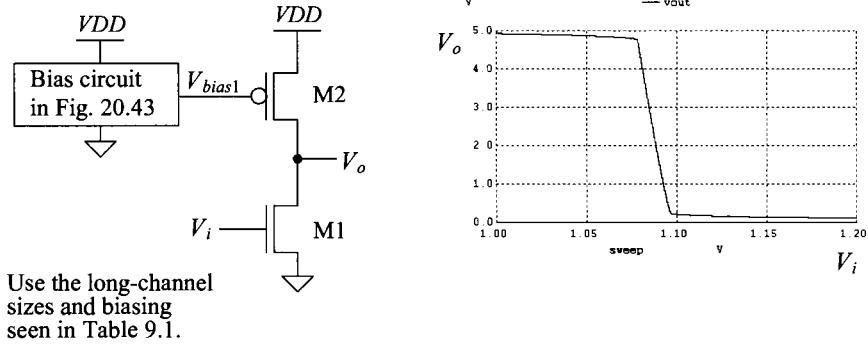


Figure 21.17 Common-source amplifier with current source load.

Class A Operation

We know from Table 9.1, and the bias circuit in Fig. 20.43, that M2 in Fig. 21.17 is biased to behave like a 20 μA current source. This means that if M1 shuts off, the maximum current we can supply to the output is 20 μA : this is important to understand. The CS amplifier in Fig. 21.17 is called a *class A* amplifier because, for proper operation, both MOSFETs are always conducting a current. In a *class B* amplifier, only one MOSFET is conducting a current at a given time. In *class AB* amplifiers (which we'll talk about later), both MOSFETs or a single MOSFET conduct a current at a given time.

Example 21.5

Estimate the maximum rate the load capacitor in Fig. 21.18 can be charged (estimate the slew rate across the load capacitor). Verify the estimate with SPICE.

The maximum current that the amplifier can source is 20 μA . The slew rate across the load is calculated using

$$I = C_L \cdot \frac{dV_{out}}{dt} \text{ or slew rate} = \frac{dV_{out}}{dt} = \frac{I}{C_L} \quad (21.38)$$

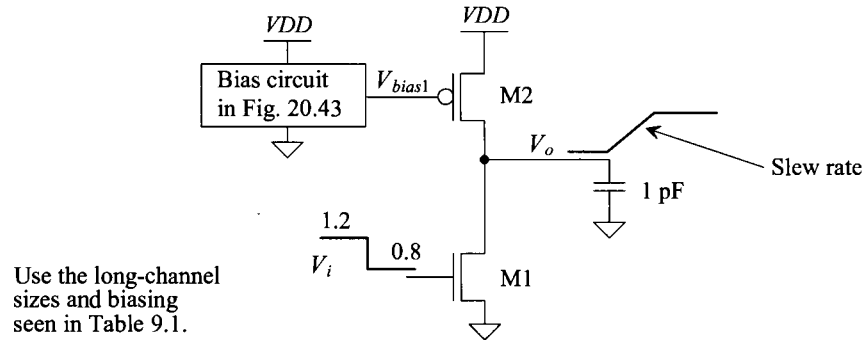


Figure 21.18 Slew rate limitations in a class A amplifier.

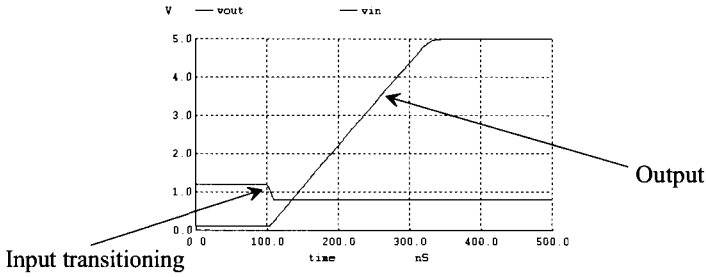


Figure 21.19 Verifying the results in Ex. 21.5

or

$$\frac{dV_{out}}{dt} = \frac{20 \mu A}{1 pF} = 20 V/\mu s$$

For the output to transition 5 V requires 250 ns. ■

Small-Signal Gain

If we follow the intuitive method discussed in the last section, Eq. (21.1), the voltage gain of a common-source amplifier is the total parallel resistance at the drain of M1 divided by the resistance in the source of M1, or

$$A_v = \frac{v_o}{v_i} = -\frac{r_{o1} || r_{o2}}{\frac{1}{g_{m1}}} = -g_{m1}(r_{o1} || r_{o2}) = -\frac{g_{m1}}{g_{o1} + g_{o2}} \quad (21.39)$$

where $r_{o1} = 1/g_{o1}$.

Open Circuit Gain

The gain of the amplifier can be increased by using a cascode current load in place of M2. The resistance looking into the drain of the cascode current source/load is much larger than the output resistance of M1. This situation is sometimes referred to as the *open circuit gain* of a common-source amplifier (see the specification in Tables 9.1 and 9.2). This specification indicates the maximum possible gain attainable with a single MOSFET (the MOSFET's load is its own output resistance r_o). The open circuit gain of M1 can be written as

$$\text{Open circuit gain} = -\frac{r_{o1}}{\frac{1}{g_{m1}}} = -g_{m1}r_{o1} = -\frac{\sqrt{2\beta_1 I_D}}{I_D \lambda_1} \quad (21.40)$$

High-Impedance and Low-Impedance Nodes

In Fig. 21.17, the output node at the drains of M1 and M2 is termed a *high-impedance node*, that is, a node with only drain connections. The effective resistance at this node to ground is $r_{o1} || r_{o2}$. A node connected to the source of a MOSFET or a MOSFET with its drain and gate connected is termed a *low-impedance node*. The small-signal resistance looking into the source of a MOSFET and the small-signal resistance of the gate-drain (diode)-connected MOSFET are both $1/g_m$. High-impedance nodes usually have a low-frequency pole associated with them that limits the speed of the amplifier.

Frequency Response

Figure 21.20 shows the AC frequency response circuit for the CS amplifier with current source load in Fig. 21.17. V_{DD} in this figure is drawn as an AC ground. The source-gate capacitance of M2 isn't shown because both sides of it are at AC ground (DC voltages), so it doesn't affect the frequency behavior of the circuit. The pole associated with the input node of the circuit is

$$f_{in} = \frac{1}{2\pi(C_{gs1} + C_{gd1}(1 + |A_v|)) \cdot R_s} \quad (21.41)$$

while the pole associated with the output node of the circuit is located at

$$f_{out} = \frac{1}{2\pi\left(C_{dg2} + C_{gd1}\left(1 + \frac{1}{|A_v|}\right)\right) \cdot r_{o1} \parallel r_{o2}} \quad (21.42)$$

where $A_v = -g_{m1} \cdot r_{o1} \parallel r_{o2}$ (see Eq. [21.39]). Using Eq. (21.42) to calculate the output pole, results in a *wrong estimate* when A_v is large. As we'll see in a moment, the fact that A_v decreases above f_{in} causes v_{in} to “flatten out” because the Miller capacitance, $C_{gd1}(1 + |A_v|)$, gets smaller. The effect of v_{in} ceasing to decrease causes the output pole to appear much larger than what is predicted by Eq. (21.42). This important phenomenon is called *pole splitting*.

The zero in the transfer function (see Eq. (21.15)) is located at

$$f_z = \frac{g_{m1}}{2\pi C_{gd1}} \quad (21.43)$$

The transfer function of the amplifier is then estimated as

$$A_v(f) = -g_{m1} \cdot (r_{o1} \parallel r_{o2}) \cdot \frac{\left(1 - j\frac{f}{f_z}\right)}{\left(1 + j\frac{f}{f_{in}}\right)\left(1 + j\frac{f}{f_{out}}\right)} \quad (21.44)$$

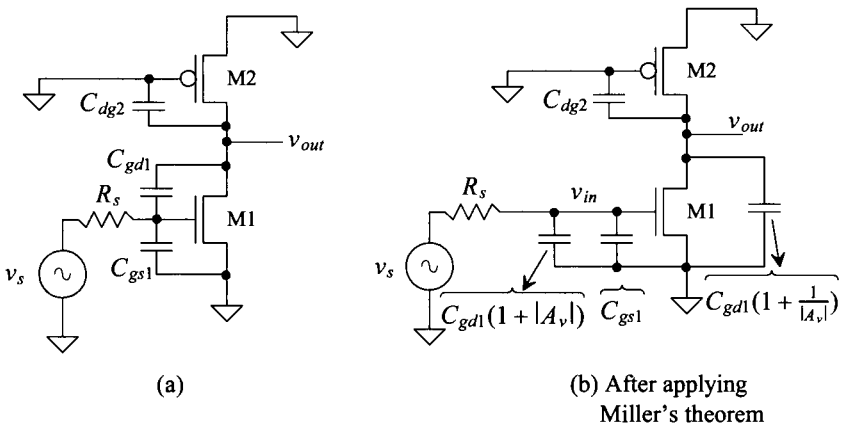


Figure 21.20 Frequency response of the CS amplifier with current source load.

We can then write the magnitude as

$$|A_v(f)| = \frac{g_{m1} \cdot (r_{o1} || r_{o2}) \cdot \sqrt{1 + \left(\frac{f}{f_z}\right)^2}}{\left(\sqrt{1 + \left(\frac{f}{f_{in}}\right)^2}\right) \cdot \left(\sqrt{1 + \left(\frac{f}{f_{out}}\right)^2}\right)} \quad (21.45)$$

and the phase shift through the amplifier as

$$\angle A_v = 180 - \tan^{-1}\left(\frac{f}{f_z}\right) - \tan^{-1}\left(\frac{f}{f_{in}}\right) - \tan^{-1}\left(\frac{f}{f_{out}}\right) \quad (21.46)$$

Example 21.6

Determine the gain, magnitude, and phase shift of the amplifier shown in Fig. 21.21. Using an AC SPICE simulation, verify your hand calculations.

For the AC small-signal analysis, the big resistor appears like an open and the big capacitor a short. These components are added to ensure that M1 is biased to sink the current supplied by M2.

The low-frequency gain of the circuit is, from Eq. (21.39) and Table 9.1,

$$|A_v| = (150 \mu A/V) \cdot (5M\Omega || 4M\Omega) = 333 V/V \rightarrow 50 \text{ dB}$$

The input capacitance is

$$C_{in} = (C_{gs1} + C_{gd1}(1 + |A_v|)) = (23.3 + 2 \cdot 333) \text{ fF} = 691 \text{ fF}! \quad (21.47)$$

The output capacitance is

$$C_{out} = \left(C_{dg2} + C_{gd1}\left(1 + \frac{1}{|A_v|}\right)\right) \approx 6 + 2 \text{ fF} = 8 \text{ fF} \quad (21.48)$$

Using Eqs. (21.41)–(21.43), we get $f_{in} = 2.3 \text{ MHz}$, $f_{out} = 9 \text{ MHz}$, and $f_z = 11.9 \text{ GHz}$. Figure 21.22 shows the simulation results. As the discussion following Eq. (21.42) indicated, the estimate for f_{out} is wrong. To determine the correct value, let's discuss pole splitting (see Eq. [21.62]). ■

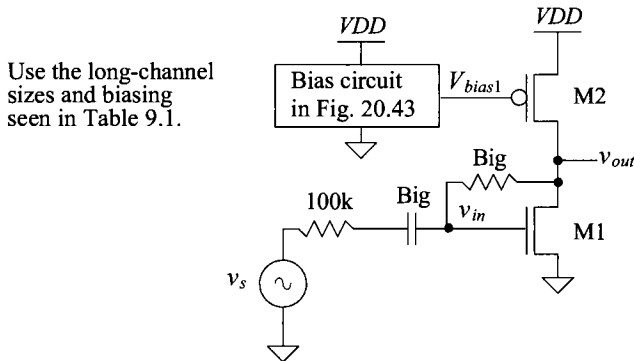


Figure 21.21 Amplifier used in Ex. 21.6.

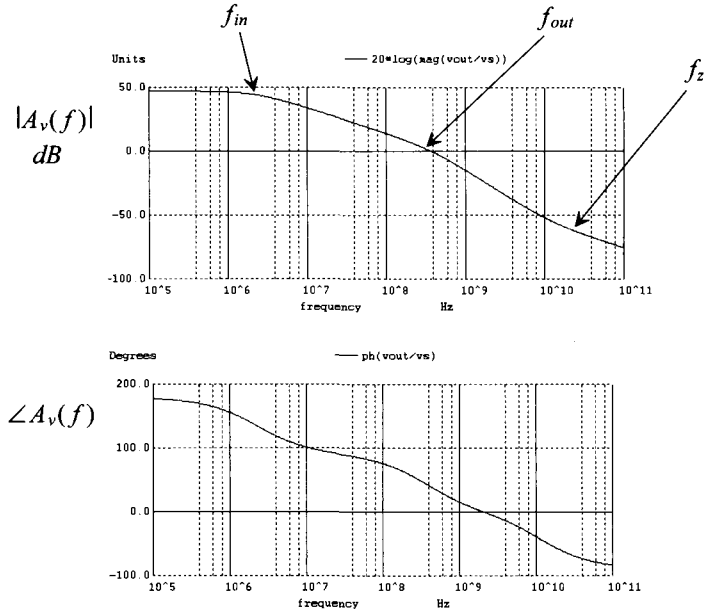


Figure 21.22 Frequency response of the amplifier in Fig. 21.21.

Pole Splitting

As discussed after Eq. (21.42), using Miller’s theorem to calculate the location of the output pole when the gain of the amplifier is large results in error. Clearly, as just demonstrated in Ex. 21.6, the calculated value of f_{out} is 9 MHz while the SPICE simulation in Fig. 21.22 shows that f_{out} is approximately 200 MHz. The source of this discrepancy can be traced to the fact that above f_{in} the gain of the amplifier, A_v , decreases causing the effective input capacitance of the amplifier to decrease. As seen in Fig. 21.23, this causes the input voltage to “flatten out” instead of continuing to fall at -20 dB/decade. The result is the second, or output pole, appears to split from (leave) the first,

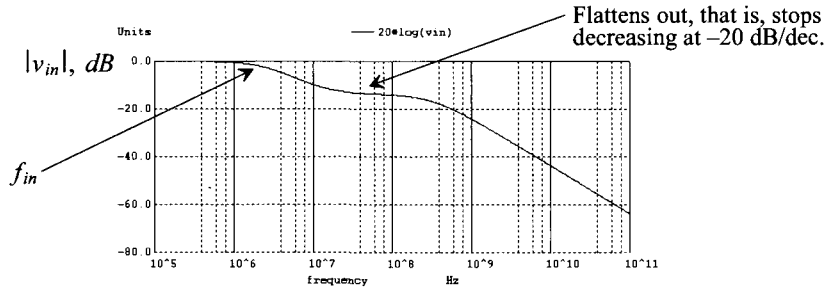


Figure 21.23 Showing how the amplifier’s, in Fig. 21.21, input voltage flattens out as the gain decreases.

or input pole, and move to a higher frequency. Note that if we resimulate the amplifier in Fig. 21.21 with the source resistance set to zero ohms (so there isn't an input pole), pole splitting isn't present and Eq. (21.42) can be used to calculate the location of the output pole.

To characterize this effect, let's use the circuit in Fig. 21.7 with the addition of a source resistance, R_s , as seen in Fig. 21.24. In this figure, $R_o = r_{o1} || r_{o2}$ and $C_o = C_{dg2}$. From Eq. (21.14), we can write

$$\frac{v_{out}}{v_{in}} = -g_{m1} R_o \cdot \frac{1 - j\omega \frac{C_{gd1}}{g_{m1}}}{1 + j\omega(C_{gd1} + C_o) \cdot R_o} \quad (21.49)$$

At the gate of M1, we can write

$$\frac{v_{in} - v_s}{R_s} + \frac{v_{in}}{1/j\omega C_{gs1}} + \frac{v_{in} - v_{out}}{1/j\omega C_{gd1}} = 0 \quad (21.50)$$

or

$$v_{in} = \frac{\frac{v_s}{R_s} + v_{out} \cdot j\omega C_{gd1}}{\frac{1}{R_s} + j\omega C_{gs1} + j\omega C_{gd1}} \quad (21.51)$$

Substituting Eq. (21.51) into Eq. (21.49) gives

$$v_{out} = -g_{m1} R_o \cdot \frac{1 - j\omega \frac{C_{gd1}}{g_{m1}}}{1 + j\omega(C_{gd1} + C_o) \cdot R_o} \cdot \frac{\frac{v_s}{R_s} + v_{out} \cdot j\omega C_{gd1}}{\frac{1}{R_s} + j\omega C_{gs1} + j\omega C_{gd1}} \quad (21.52)$$

or, with $s = j\omega$,

$$\frac{v_{out}}{v_s} = \frac{-g_{m1} R_o \cdot \left(1 - s \frac{C_{gd1}}{g_{m1}}\right)}{s^2 [R_o R_s (C_{gd1} C_{gs1} + C_o C_{gs1} + C_o C_{gd1})] + s [(C_{gd1} + C_o) R_o + (C_{gs1} + C_{gd1}) R_s + C_{gd1} g_{m1} R_o R_s] + 1} \quad (21.53)$$

The zero is still, as indicated in Eqs. (21.15) and (21.43), located at

$$f_z = \frac{g_{m1}}{2\pi C_{gd1}} \quad (21.54)$$

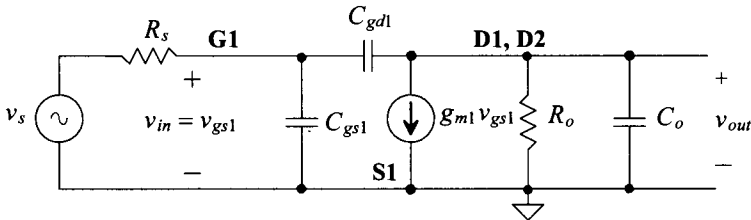


Figure 21.24 Circuit used to calculate the transfer function of the amplifier in Fig. 21.21.

At low frequencies the denominator is approximately (the coefficient of the s^2 term is small)

$$1 + j\omega \cdot [(C_{gd1} + C_o)R_o + (C_{gs1} + C_{gd1})R_s + C_{gd1}g_{m1}R_oR_s] \quad (21.55)$$

and so the low-frequency pole, using $|A_v| = g_{m1}(r_{o1} || r_{o2}) = g_{m1}R_o$, $C_o = C_{dg2}$, is located at

$$f_1 \approx \frac{1}{2\pi[(C_{gd1} + C_{dg2}) \cdot r_{o1} || r_{o2} + (C_{gs1} + C_{gd1}(1 + |A_v|)) \cdot R_s]} \quad (21.56)$$

If $C_{gd1}(1 + |A_v|)$ is much larger than the other capacitances,

$$f_1 \approx \frac{1}{2\pi C_{gd1}(1 + |A_v|) \cdot R_s} \quad (21.57)$$

Notice the similarity to Eq. (21.41), where the pole is associated with the input of the amplifier. Further notice that if we had used this result in Ex. 21.6, the input pole would have changed very little. Also notice that if $R_s = 0$, then

$$f_1 \approx \frac{1}{2\pi(C_{gd1} + C_{dg2}) \cdot r_{o1} || r_{o2}} \quad (21.58)$$

Notice the similarity to Eq. (21.42), where the pole is associated with the output of the amplifier. To determine the location of the second root of the denominator (the second pole location, f_2), let's factor Eq. (21.55) from the denominator of Eq. (21.53) or

$$[1 + s[(C_{gd1} + C_o)R_o + (C_{gs1} + C_{gd1})R_s + C_{gd1}g_{m1}R_oR_s]] \cdot \left(1 + \frac{s^2[R_oR_s(C_{gd1}C_{gs1} + C_oC_{gs1} + C_oC_{gd1})]}{s[(C_{gd1} + C_o)R_o + (C_{gs1} + C_{gd1})R_s + C_{gd1}g_{m1}R_oR_s] + 1} \right) \quad (21.59)$$

where this equation is in the form

$$\left(1 + j \cdot \frac{f}{f_1} \right) \cdot \left(1 + j \cdot \frac{f}{f_2} \right) \quad (21.60)$$

Note again that if $R_s = 0$, the second term in Eq. (21.59) goes to unity. Looking at the second term of Eq. (21.59), let's divide the numerator and denominator by sR_oR_s

$$1 + j \cdot \frac{2\pi f \cdot (C_{gd1}C_{gs1} + C_oC_{gs1} + C_oC_{gd1})}{(C_{gd1} + C_o)/R_s + (C_{gs1} + C_{gd1})/R_o + C_{gd1}g_{m1} + 1/sR_oR_s} \quad (21.61)$$

In any practical MOSFET amplifier, $g_m \gg 1/r_o$ (if not, then the magnitude of the open circuit gain, Eq. (21.40), is too small for the MOSFET to be of practical value as an amplifying device). Also, the R_s is assumed to be large (if not, then f_2 is not used), that is, the amplifier's transfer function has a single-pole response, where the pole is associated with the output of the amplifier as seen in Eq. (21.58). We can therefore write

$$f_2 \approx \frac{g_{m1}C_{gd1}}{2\pi \cdot (C_{gd1}C_{gs1} + C_oC_{gs1} + C_oC_{gd1})} \quad (21.62)$$

Calculating the location of the second pole in Ex. 21.6 with this equation results in $f_2 = 240$ MHz (which is very close to the simulated location).

Why the name pole splitting? If we increase the effective size of C_{gd1} by placing a capacitor, C_c , from the amplifier's input to its output (so that C_c is in parallel with C_{gd1} , that is, the effective value is $C_c + C_{gd1}$), then, as Eq. (21.56) shows, the low-frequency location of the pole, f_1 , decreases. At the same time, increasing the effective size of C_{gd1} causes the location of the high-frequency pole f_2 , as seen in Eq. (21.62), to increase. Thus the name *pole splitting*.

Pole Splitting Summary

Because of the ubiquity of pole splitting in CMOS amplifier design, let's summarize our discussion using a generic model, Fig. 21.25. In terms of the parameters seen in this figure the location of the zero is now

$$f_z = \frac{g_{m2}}{2\pi \cdot C_c} \quad (21.63)$$

The first pole is located at

$$f_1 \approx \frac{1}{2\pi[(C_c + C_2) \cdot R_2 + (C_1 + C_c(1 + g_{m2}R_2)) \cdot R_1]} \quad (21.64)$$

For large gain, $g_{m2}R_2$, we can simplify this to

$$f_1 \approx \frac{1}{2\pi g_{m2}R_2R_1C_c} \quad (21.65)$$

The second pole is located at

$$f_2 \approx \frac{g_{m2}C_c}{2\pi \cdot (C_cC_1 + C_1C_2 + C_cC_2)} \quad (21.66)$$

The transfer function is then written as

$$A_v(f) = \frac{v_{out}(f)}{v_s(f)} = g_{m1}R_1g_{m2}R_2 \cdot \frac{\left(1 - j \cdot \frac{f}{f_z}\right)}{\left(1 + j \cdot \frac{f}{f_1}\right) \cdot \left(1 + j \cdot \frac{f}{f_2}\right)} \quad (21.67)$$

where $A_v = g_{m1}R_1g_{m2}R_2$ (the gain is now positive because of the defined direction of the current source in Fig. 21.25).

Again note that if C_c is increased, f_1 moves downwards and f_2 moves upwards (pole splitting). Also note that if the capacitor, C_c , is made large (quite common), then f_1 is much lower than the location of the other pole or the zero (the pole associated with f_1 is

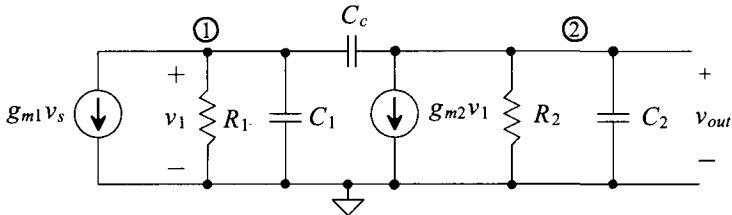


Figure 21.25 Generic model used to estimate bandwidth in a CMOS amplifier.

said to be the *dominant pole*). The amplifier's transfer function, with the help of Eq. (21.65), can be written as

$$A_v(f) = \frac{v_{out}(f)}{v_s(f)} \approx \frac{g_{m1}R_1g_{m2}R_2}{\left(1+j\cdot\frac{f}{f_1}\right)} = \frac{g_{m1}R_1g_{m2}R_2}{(1+j\cdot 2\pi f\cdot g_{m2}R_2R_1C_c)} \quad (21.68)$$

This equation is in the form

$$A_v(f) = \frac{v_{out}(f)}{v_s(f)} = \frac{A_{DC}}{1+j\cdot\frac{f}{f_{3dB}}} \quad (21.69)$$

where

$$A_{DC} = g_{m1}R_1g_{m2}R_2 \text{ and } f_{3dB} = \frac{1}{2\pi g_{m2}R_2R_1C_c} \quad (21.70)$$

The magnitude and phase responses of a generic CMOS amplifier are seen in Fig. 21.26. Note that at frequencies much larger than the amplifier's 3 dB frequency, that is, $f/f_{3dB} \gg 1$, Eq. (21.68) can be written as

$$A_v(f) = \left| \frac{v_{out}(f)}{v_s(f)} \right| \approx \frac{g_{m1}}{2\pi f\cdot C_c} \quad (21.71)$$

To determine the frequency, f_{un} , where the transfer function is unity, we set Eq. (21.71) to one and solve to yield

$$f_{un} = \frac{g_{m1}}{2\pi C_c} \quad (21.72)$$

We've derived some very useful equations, so let's apply them in some examples.

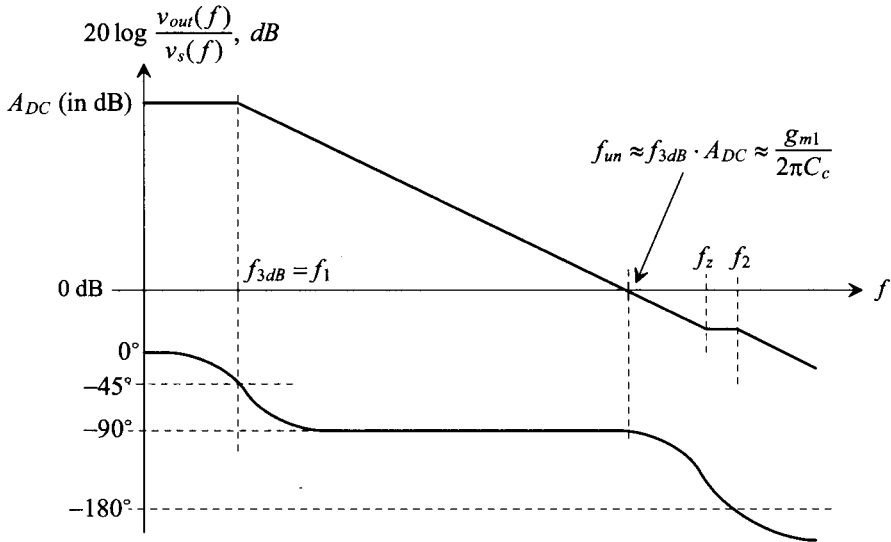


Figure 21.26 Magnitude and phase responses of a generic CMOS amplifier.